

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1

- Problem 1** (a) Let $f(x) = \frac{x-1}{x^2-5x}$. Then $f(2) = -\frac{1}{6}$ and $f(6) = \frac{5}{6}$, but there is no value of c between 2 and 6 for which $f(c) = 0$. Does this fact violate the Intermediate Value Theorem?
- (b) True or False: At some time since you were born your weight in pounds exactly equaled your height in inches.

Problem 2

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Problem 2

Problem 2 For the following function g defined by

$$g(t) = \begin{cases} 5t + 7 & \text{if } t < -3 \\ \frac{(t-1)(t+2)}{t+2} & \text{if } -3 \leq t < 1 \text{ and } t \neq -2 \\ 4 \ln t & \text{if } t \geq 1 \end{cases}$$

find the **Intervals of Continuity**.

(**Important Note:** Write your answer as a list of intervals, with each interval separated by a comma.)

Problem 3

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Problem 3

Problem 3 Determine the value of a constant b for which f is continuous at 0. **EXPLAIN.**

$$f(x) = \begin{cases} \frac{2x+b}{x-5} & \text{if } x < 0 \\ \frac{x+16}{x^2-16} & \text{if } x \geq 0 \end{cases}$$

Problem 4

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Problem 4

Problem 4 Use the Intermediate value theorem to find an interval in which you can guarantee that there is a solution to the equation $x^3 = x + \sin(x) + 1$. **EXPLAIN.** (Do not use a graphing device or calculator to solve this problem!)

Problem 5

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Problem 5 (a) True or False: If f and g are two functions defined on $(-1, 1)$, and if $\lim_{x \rightarrow 0} g(x) = 0$, then it must be true that $\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = 0$.

(b) True or False: If f is continuous on $(-1, 1)$, and if $f(0) = 10$ and $\lim_{x \rightarrow 0} g(x) = 2$, then

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 5.$$

(c) True or False: If f is continuous on $[1, 3]$, and if $f(1) = 0$ and $f(3) = 4$, then the equation $f(x) = \pi$ has a solution in $(1, 3)$.

(d) True or False: Let f be a positive function with vertical asymptote $x = 5$. Then

$$\lim_{x \rightarrow 5} f(x) = \infty.$$

Problem 6

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Problem 6 Let

$$h(u) = \begin{cases} \frac{u^2 - 5u + 4}{u - 4} & \text{if } u < 4 \\ \frac{-\sqrt{u + 4}}{u - 6} & \text{if } u \geq 4, u \neq 6. \end{cases}$$

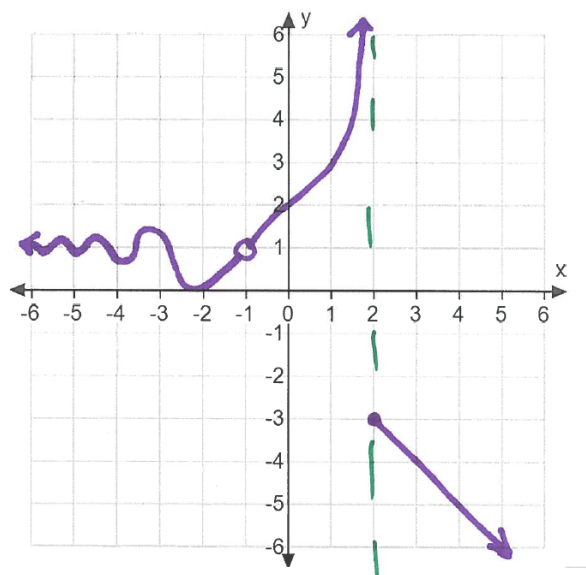
- (a) What is the domain of h ?
- (b) Find all vertical asymptotes of h . **EXPLAIN.**
- (c) Find all horizontal asymptotes of h . **EXPLAIN.**
- (d) List the **Intervals of Continuity** for the function h .

Problem 7

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Problem 7

Problem 7 Use the graph of f to answer the questions below.



- (a) State the domain of f .
- (b) Find the following values or state "does not exist":
 - (i) $\lim_{x \rightarrow 2^-} f(x) =$
 - (ii) $\lim_{x \rightarrow 2^+} f(x) =$
 - (iii) $\lim_{x \rightarrow 2} f(x) =$
 - (iv) $\lim_{x \rightarrow -1} f(x) =$
 - (v) $\lim_{x \rightarrow 3} f(x) =$
 - (vi) $\lim_{x \rightarrow -\infty} f(x) =$
 - (vii) $f(-1) =$
- (c) State the equation of any vertical asymptotes.
- (d) State the equation of any horizontal asymptotes.
- (e) Find the Intervals of Continuity.

Problem 8

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Problem 8

Problem 8 Suppose a taxi ride costs \$7.50 for the first mile (or any part of the first mile), plus an additional \$1.00 for each additional mile (or any part of a mile).

- (a) Graph the function $c = f(t)$ that gives the cost of a taxi ride for t miles, for $0 \leq t \leq 5$.
- (b) Evaluate $\lim_{t \rightarrow 2.9} f(t)$
- (c) Evaluate $\lim_{t \rightarrow 3^-} f(t)$ and $\lim_{t \rightarrow 3^+} f(t)$
- (d) Interpret the meaning of the limits in part (c).
- (e) On what intervals is the function c continuous? Explain.

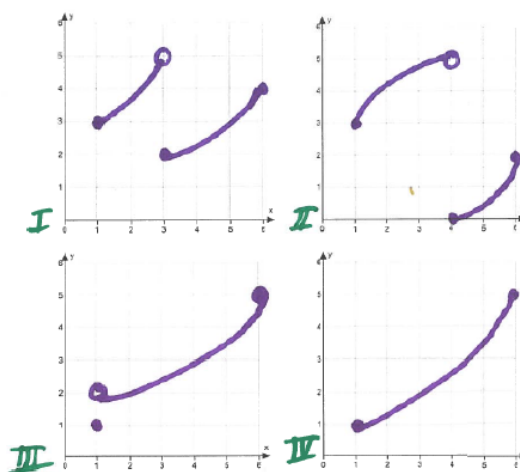
Problem 9

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Problem 9 (a) Given function f on an interval $[a, b]$:

- (i) What are the conditions of the Intermediate Value Theorem?
- (ii) What is the conclusion of the Intermediate Value Theorem?

(b) Given the four functions on the interval $[1, 6]$, answer the questions below.



- (i) For each of the functions I through IV, indicate $f(1)$ and $f(6)$. Then mark the interval of all numbers strictly between $f(1)$ and $f(6)$, on the y-axis.
- (ii) For each of the functions I through IV, write an interval of all numbers strictly between $f(1)$ and $f(6)$
- (iii) List the functions that satisfy the conditions of the Intermediate Value Theorem on $[1, 6]$
- (iv) For which of the functions is the following statement true: For any number L strictly between $f(1)$ and $f(6)$, there exists a number c in $(1, 6)$ satisfying $f(c) = L$.
- (v) Does the function III satisfy the conclusion of the Intermediate Value Theorem? Why or why not?